

Dennis Delali Kwesi Wayo

Quantum Error Correction

& Quantum Architecture Scientist — Scientific Software Engineer

Astana, Kazakhstan | +7 771 414 0389

✉ dwayo3@gatech.edu  LinkedIn  GitHub

Summary

Research engineer building fault-tolerant quantum software for photonic and hybrid architectures. Experience spans decoder benchmarking, architecture-level simulation, and reproducible scientific computing pipelines in Python, Rust, and C++. Built end-to-end QEC tooling across hardware-to-decoder interfaces, with emphasis on decoder comparability, reproducibility, and architecture-level performance analysis. Seeking research-engineering roles in quantum architecture and fault-tolerant systems where scalable simulation and decoder infrastructure inform hardware and compiler decisions.

Core Skills

Programming: Python, Swift, Rust, C++, Julia, Bash

QEC & Architecture: Surface-code threshold workflows, decoder benchmarking (BP/MWPM/UF), GKP digitization, photonic architecture modeling

Quantum Tooling: PennyLane, Qiskit, Mitiq, Qibo, QuTiP

Scientific Computing: Numerical linear algebra, HPC Linux workflows, parallel simulation, reproducibility and artifact tracking

Software Engineering: Git, code review, CI/CD (GitHub Actions), testing (pytest), Docker, REST-style API development

ML for Science: PyTorch, TensorFlow, JAX, differentiable simulation

Research & Engineering Experience

LiDMaS+ — Logical Decoder Benchmarking Engine (C++/Rust) 2026–Present

- Built replay-driven QEC benchmarking pipelines with normalized decoder I/O to compare BP, MWPM, UF, and neural-style adapters on identical syndrome streams.
- Added diagnostics for syndrome satisfaction, residual nonzero rate, flip sparsity, and decoder stability across matched datasets.
- Implemented deterministic reproducibility checks (SHA-256 artifact matching) and Rust backend APIs for provider onboarding and run tracking.

SchroSIM — Hardware-Agnostic Quantum Photonic Simulator (Swift/Rust) 2024–Present

- Built SchroSIM as a layered photonic-circuit platform separating circuit authoring, compilation, numerical execution, and hardware mapping for independent optimization.
- Implemented hybrid numerical pathways: efficient symplectic-state propagation for Gaussian blocks and higher-fidelity Fock-space/tensor-network style routines for non-Gaussian blocks.
- Developed Swift/Rust kernels and architecture-level interfaces for hardware-aware CV simulation, including GKP logical-state studies under loss and dephasing.

Research Assistant — Nazarbayev University 2021–Present

- Developed multiphysics and quantum-adjacent simulation workflows with finite-difference methods and HPC execution pipelines.
- Built DFT/TDDFT simulation stacks (GPAW, ASE) for photonic material modeling and spectral analysis.
- Integrated ML-assisted scientific workflows with physics-based simulation and reproducible experiment tracking.

Education

- PhD, Computer Science (Extended)** 2029
TU Bergakademie Freiberg, Faculty of Mathematics and Computer Science, Germany
Research Focus: Parallel Knowledge-Graph and AI-Augmented Compiler Architectures for Scalable Logical-GKP Quantum Error Correction in Continuous-Variable Quantum Systems
- PhD, Chemical Engineering** 2026
Universiti Malaysia Pahang, FTKKP, Malaysia
Research Focus: Quantum photonic-induced solid-state electrolyte for CO₂ reduction
- M.S., Computer Science (OMSCS)** 2027
Georgia Institute of Technology, College of Computing, Atlanta, USA
Speciality: Computing Systems; Algorithms, quantum computing/hardware, Machine Learning, NLP, HPC, High Performance Computing Architecture, Software Architecture & Design, Embedded systems optimisation, & Advanced Operating System

Selected Publications & Preprints

- Wayo, D.D.K. et al. (2026). A unified hardware-to-decoder architecture for hybrid continuous-variable and discrete-variable quantum error correction in LiDMaS+. *arXiv* 2604.15389.
- Wayo, D.D.K. & Groppe, S. (2026). RaCS: near-zero-error classical data encoding on photonic quantum processors. *Fortschritte der Physik*, 74(4), e70095.
- Wayo, D.D.K. et al. (2026). Decoder dependence in surface-code threshold estimation with native GKP digitization. *arXiv* 2603.25757.
- Wayo, D.D.K. et al. (2026). Decoder performance in hybrid CV-discrete surface-code threshold estimation using LiDMaS+. *arXiv* 2603.06730.
- Wayo, D.D.K. (2025). Simulation of ultrafast photonic circuits via nonlinear Schrödinger dynamics and quantum detector modeling. *Optical and Quantum Electronics*.
- Wayo, D.D.K. et al. (2025). Atomistic modeling of rare earth ions in photonic materials. *Luminescence*.
- Wayo, D.D.K. (2025). Ensembles of graph and physics-informed machine learning for scientific modeling. *Archives of Computational Methods in Engineering*.

Honors & Service

- Womanium & WISER Quantum Scholar (2025); IBM Qiskit Advocate and QAMP Mentor
- Reviewer: Journal of Open Source Software, Scientific Reports, Nanoscale Advances